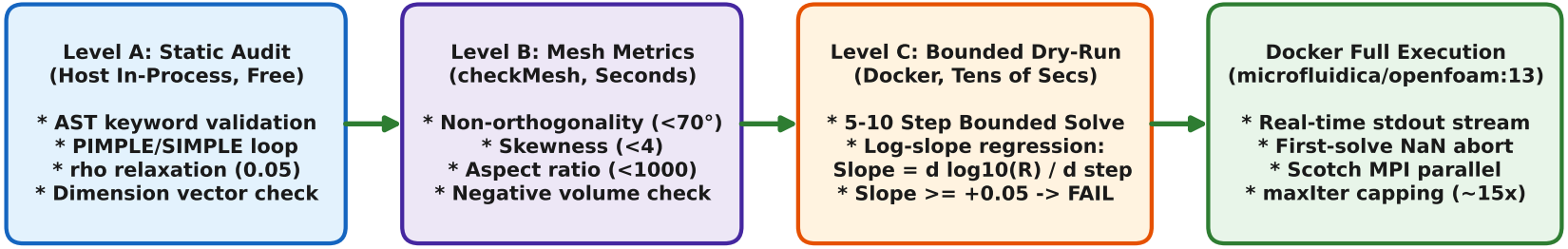
Defense-in-Depth: 4-Tier Automated Validation Ladder & Preflight Gate



Key Engineering Impact of the Tiered Validation Ladder:

1. Levels A & C catch >90% of erroneous cases before long container runs -> Saves hours of wasted compute.
2. Real-time stdout NaN abort terminates diverging containers in ~2 seconds instead of waiting for 600s timeout.